Fleet Overview

Global ID: FP-8820

System Live

+ Register Motor

MOTORS ONLINE	HEALTHY	WARNING	CRITICAL	ENERGY TODAY	$ SAVED
09/10 (90%)	06	02	01	128.4 kWh	$4,290

Main Production Line

Pump Motor A1 — HEALTHY
ID: M-10042

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
1,747	42.8 °C	1.2 mm/s	45.3 A	98%

Conveyor B-Left — WARNING
ID: M-22091

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
1,419	65.1 °C	4.8 mm/s	58.5 A	72%

Fan Unit Main — CRITICAL
ID: M-44823

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
2,098	88.3 °C	12.5 mm/s	72.3 A	34%

Mill Drive 01 — HEALTHY
ID: M-10112

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
852	38.1 °C	0.8 mm/s	32.6 A	100%

Mixer X-4 — OFFLINE
ID: M-99201

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
—	—	—	—	—

Conveyor A-Right — HEALTHY
ID: M-22092

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
1,414	44.2 °C	1.5 mm/s	42.8 A	92%

Compressor 12 — WARNING
ID: M-55001

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
2,998	72.3 °C	5.2 mm/s	122.5 A	68%

Pump Motor A2 — HEALTHY
ID: M-10043

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
1,742	40.5 °C	1.1 mm/s	44.8 A	96%

Ventilation L1 — HEALTHY
ID: M-88210

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
1,201	32.6 °C	0.6 mm/s	18.6 A	99%

Mill Drive 02 — HEALTHY
ID: M-10113

RPM	TEMP	VIBRATION	CURRENT	HEALTH SCORE
850	39.1 °C	0.9 mm/s	31.9 A	97%

EXPO MODE

Generating synthetic data for 10 motors.



Marcus Chen
Maintenance Lead

Pump Motor A1

HEALTHY

Uptime This Month
99.8%

Control Panel

AI Predictive Analysis

"Vibration at 2x RPM up 40% over 3 days, likely shaft misalignment. Bearing temperature remains stable within nominal range. Recommend visual inspection of couplings within the next 15 days."

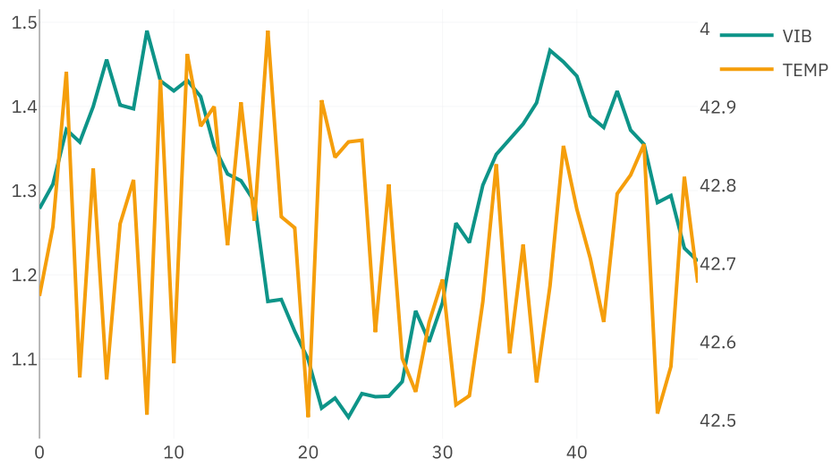
EST. FAILURE	CONFIDENCE	ANOMALY SCORE	MODEL STATUS
42 Days	94%	0.12	Optimal

Fault Probability

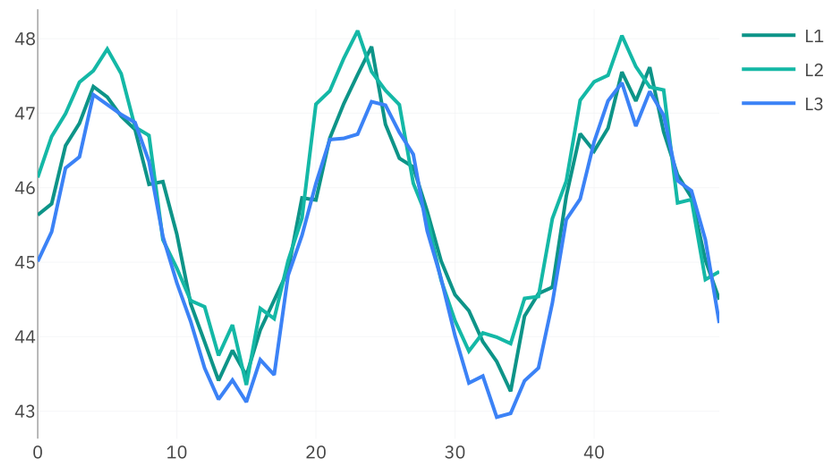
Imbalance	8%
Misalignment	62%
Bearing Wear	12%
Mechanical Looseness	4%

OVERALL HEALTH SCORE
98

Vibration & Temperature (Live)

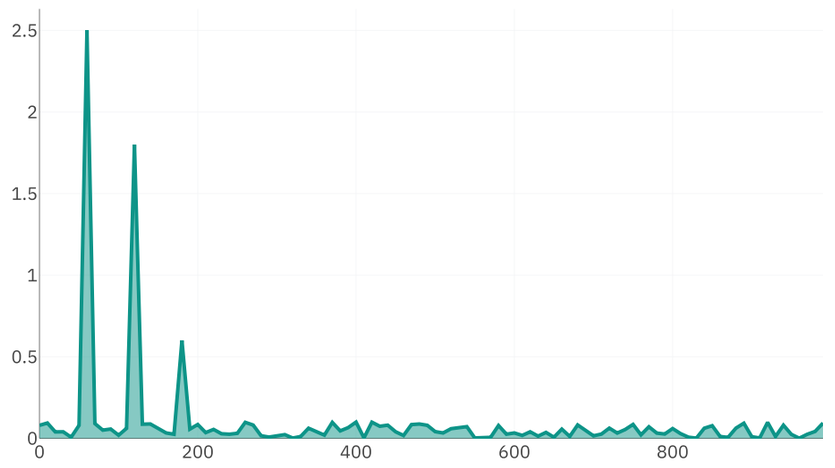


Current (Amps) - Phase Analysis

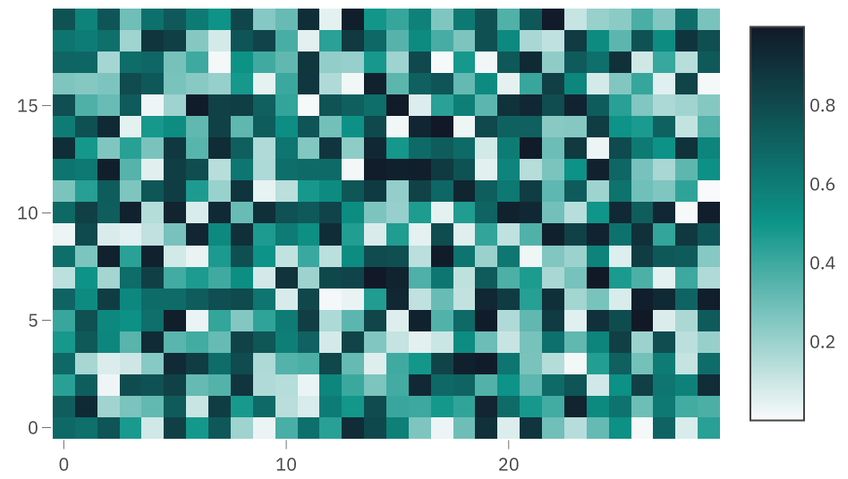


FFT Spectrum Analysis

Markers at 1x, 2x, 3x RPM



Frequency Waterfall (30 min)



Recent Events & Command Log

Predictive Warning Triggered — 10:42 AM
AI Model detected 40% increase in 2x RPM vibration peak. Misalignment probability increased to 62%.

Scheduled Maintenance Check — Yesterday, 02:15 PM
Technician (ID: 449) confirmed external coupling bolts were within torque specs.

Manual Start Command — Yesterday, 06:00 AM
Operator Marcus Chen initiated process cycle.

Control Panel

Safety Mode Active

EMERGENCY STOP ALL

FLEET STATUS	MOTORS RUNNING	ACTIVE MODE	SAFETY INTERLOCKS
Operational	09 / 10	AI-Assisted	Engaged

MOTOR UNIT	STATUS	MODE	SPEED SETPOINT	ACTIONS
Pump Motor A1 ID: M-10042	Running	AI-Assisted	1,745 RPM	
Conveyor B-Left ID: M-22091	Degraded	Manual	800 RPM	
Fan Unit Main ID: M-44823	ALARM	Manual	0 RPM	RESET ALARM
Mill Drive 01 ID: M-10112	Running	AI-Assisted	1,200 RPM	
Pump Motor A2 ID: M-10043	Running	Scheduled	1,740 RPM	

Control Audit Log

10:45:02	Marcus Chen modified Speed Setpoint for **Pump Motor A1** (1,700 -> 1,745 RPM)	Success
10:42:15	AI Agent initiated Safe Derate on **Fan Unit Main** due to vibration alarm.	SYSTEM OVERRIDE
09:15:00	Scheduled task: Shift Start Command executed globally.	Auto

ROI & Cost Savings Analysis

Reporting Period: Last 12 Months

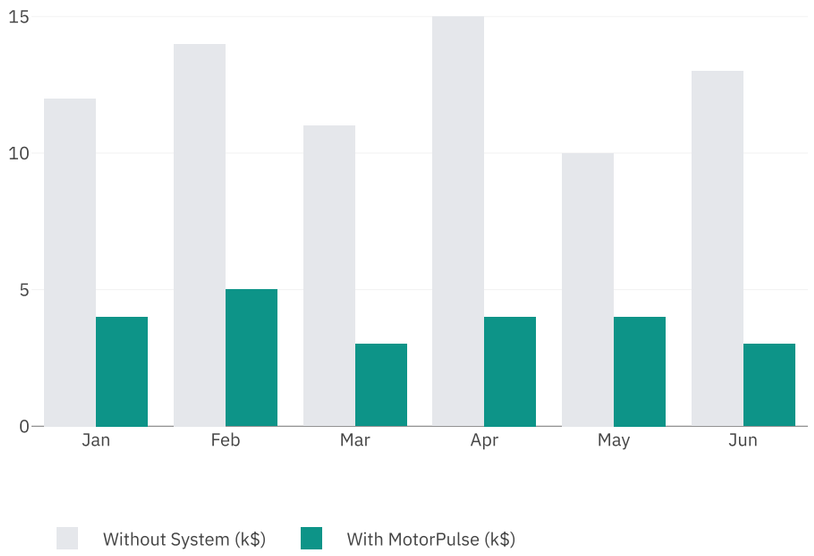
TOTAL AVOIDED LOSS
$242,500
↑ 12% vs last period

ENERGY WASTE SAVED
$18,240
Approx. 42.5 MWh prevented

AVOIDED DOWNTIME
156 Hours
8 Early fault detections

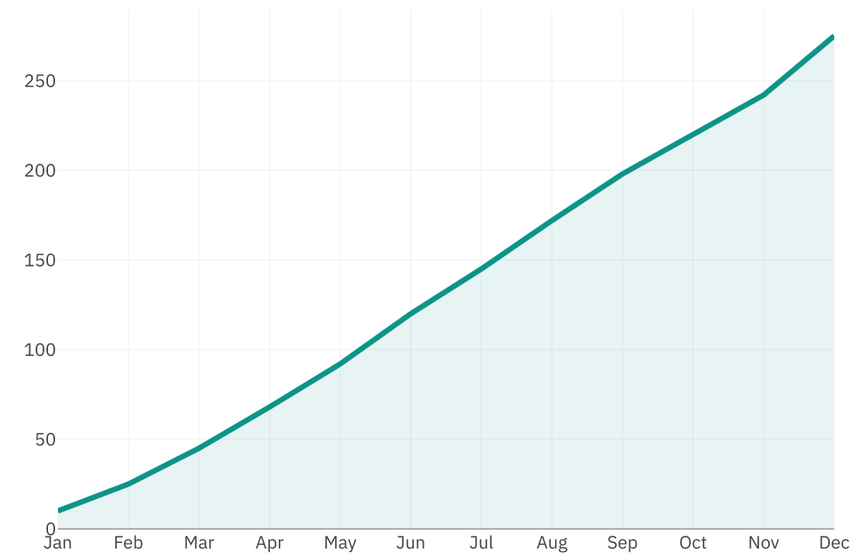
SYSTEM ROI SCORE
4.8x
Payback period: 6 months

Maintenance Costs: Planned vs Unplanned



Cumulative Savings (USD)

YEARLY PROJECTION



Savings Breakdown by Category

DOWNTIME PREVENTION
$185,000
76% of total

LABOR OPTIMIZATION
$32,000
13% of total



ROI Calculator Inputs

DOWNTIME COST / HR
$ 1500

PLANNED REPAIR COST
$ 850

MOTOR REPLACEMENT COST
$ 12000

ENERGY TARIFF (PER KWH)
$ 0.12

UNPLANNED REPAIR COST
$ 4500

TECH HOURLY RATE
$ 85

ESTIMATED ANNUAL SAVINGS
$312,000

Update Forecast

Alerts & Maintenance

Threshold Settings | Create Manual Work Order

Active Alerts (3)

Acknowledge All

CRITICAL: High Vibration Alarm — 12m ago

Motor: Fan Unit Main (ID: M-44823)

Vibration RMS exceeded 12.5mm/s threshold. AI detects 88% probability of catastrophic bearing failure within 48 hours.

ACKNOWLEDGE | GENERATE WORK ORDER

WARNING: Misalignment Detected — 1h ago

Motor: Conveyor B-Left (ID: M-22091)

Harmonic peak at 2x RPM is trending upwards (+40% in 3 days).

ACKNOWLEDGE | VIEW DIAGNOSTICS

Condition-Based Maintenance Calendar

AI DRIVEN · FIXED

MON	TUE	WED	THU	FRI	SAT	SUN
25	26	27 A1 Coupling Check	28	29 FAN-M Bearing Replace	30	31

AI Recommended Actions

SHIFT NEEDED

Move **Compressor 12** lubrication from Dec 15 to Nov 30 due to increased thermal cycling.

ACCEPT SHIFT

INVENTORY ALERT

SKF-6205 Bearings (x2) required for upcoming **Fan Unit Main** repair. Only 1 in stock.

ORDER PARTS

Maintenance History

Export Report

DATE	ASSET	TYPE	TECHNICIAN	STATUS
Oct 24, 2023	**Mill Drive 01**	Calibration	S. Jenkins	COMPLETED
Oct 22, 2023	**Pump Motor A2**	Routine Lube	M. Chen	COMPLETED

Energy Consumption & Efficiency

FLEET AVG EFFICIENCY 94.2%
ESTIMATED DAILY COST $84.12

TOTAL POWER (LIVE)
18.5 kW

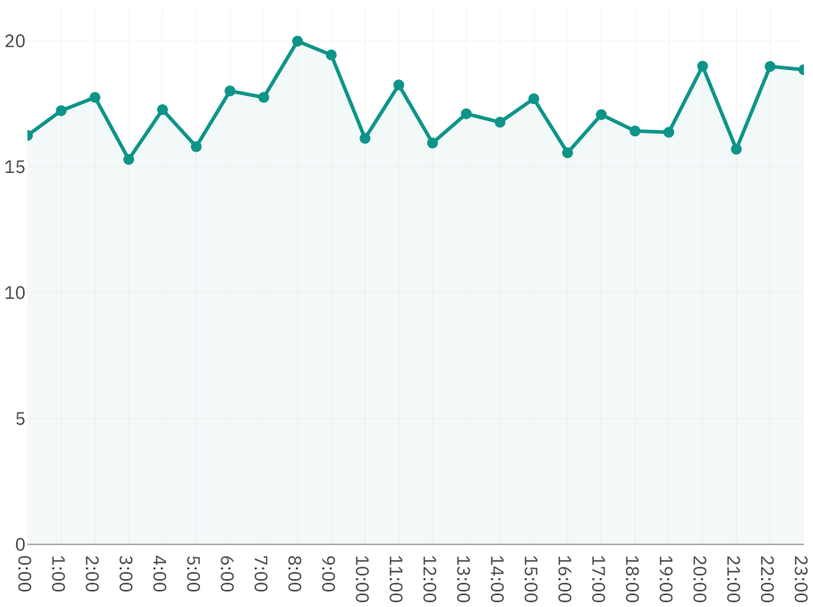
CONSUMPTION TODAY
128.4 kWh

CARBON FOOTPRINT
52.1 kg CO_2

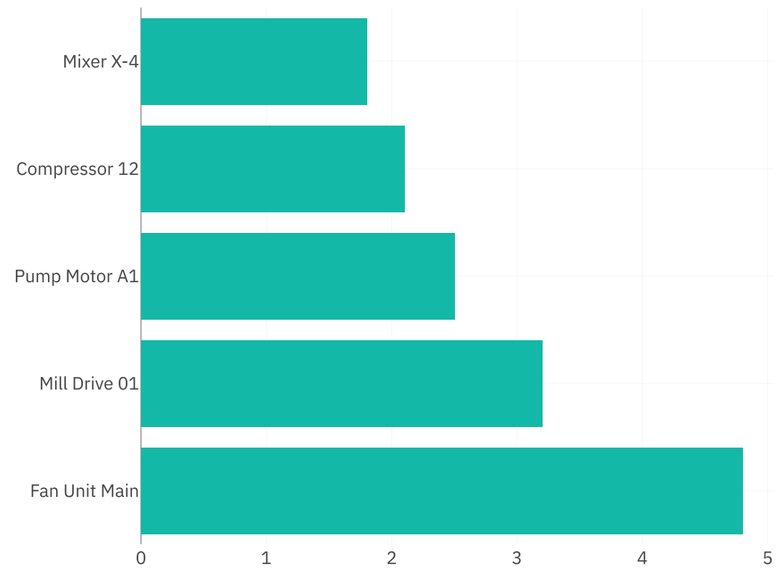
EFFICIENCY DRIFT
-1.2%
Average across fleet

Fleet Power Demand (24h)

LIVE UPDATES



Top Consuming Motors



Efficiency Drift Analysis

A degrading motor draws more current for the same work. AI correlates drift with mechanical faults.

MOTOR UNIT	POWER (KW)	EFFICIENCY	DAILY COST	AI DIAGNOSIS
Pump Motor A1	2.45	**94.2%**	**$7.06**	*Early-stage misalignment overhead*
Fan Unit Main	4.82	**82.4%**	**$13.88**	**CRITICAL BEARING FRICTION**
Mill Drive 01	3.12	**98.1%**	**$8.98**	*Normal (Healthy Baseline)*

AI-Optimized Efficiency Impact

By identifying efficiency drift early, you've prevented an estimated **$1,240** in wasted energy this month. Your carbon footprint has been reduced by **450 kg CO_2** compared to unmonitored operation.

Download Sustainability Report